

# ISI DIAGNOSTIC ENGINE TESTING TEMPLATE — Manufacturing

*(Clean, structured, ready for Cursor, and aligned with your internal/external separation rules.)*

## ENGINE 2 — SUMMARY (Client-Facing Description)

**Purpose:** A simple, high-level explanation of what Engine 2 does **from the client's perspective**, without exposing internal logic.

**Client-Facing Summary:** Engine 2 evaluates the stability, consistency, and repeatability of your production processes. It identifies where variation is occurring, how it impacts quality, and what upstream or downstream factors may be contributing to instability. The engine highlights patterns, trends, and anomalies that help determine whether the issue is related to materials, equipment, environment, or operator methods.

### What the Client Sees:

- A visual stability score
- Key variation indicators
- High-level trend flags
- A summary of potential contributing factors
- A simple explanation of what the variation means for throughput, quality, and reliability

### What the Client Does NOT See:

- SPC logic
- Capability calculations
- DOE triggers
- Measurement system validation logic
- Root cause inference engine
- Internal scoring algorithms
- Constraint mapping
- Hidden diagnostic pathways

## Manufacturing — INTERNAL LOGIC (Consultant-Only / Cursor Internal)

*(This is the intelligence layer that Cursor uses but is never shown to the client.)*

### **Internal Diagnostic Functions:**

- SPC baseline calculation
- Cp/Cpk capability scoring
- Variation clustering (material, machine, method, environment)
- DOE trigger conditions
- Measurement system confidence scoring
- Environmental correlation mapping
- Tooling wear drift detection
- Hidden bottleneck influence analysis
- Upstream/downstream propagation modeling

### **Internal Questions Engine 2 Answers:**

- Is the process stable?
- Is the process capable?
- Where is variation originating?
- Is the variation random or systemic?
- Is the variation operator-driven or equipment-driven?
- Is the variation influenced by environment?
- Does variation correlate with throughput loss?
- Does variation correlate with defect spikes?
- Does variation correlate with downtime events?

# **Manufacturing — STRUCTURED TESTING TEMPLATE (For Your Local Testing at localhost)**

Use this exact structure to document Engine 2's behavior as you test it locally.

## **1. Test Environment Setup**

**URL tested:** <http://localhost:8080/engine-testing/index.html>

**Engine selected:** Engine 2 — Process Stability & Variation Analysis

**Browser behavior:** (Describe what you see — layout, load behavior, UI elements, responsiveness.)

## **2. Input Conditions Tested**

List each input scenario you tested:

Example:

- Scenario A: Stable process, low variation
- Scenario B: High variation, random spikes
- Scenario C: Operator-driven variation
- Scenario D: Material lot-driven variation
- Scenario E: Environmental drift
- Scenario F: Tooling wear progression
- Scenario G: Mixed variation sources

### **3. Expected Output (Based on Engine 2 Logic)**

Document what the engine *should* output for each scenario:

- Stability score
- Variation flags
- Trend indicators
- High-level summary
- Suggested next diagnostic step (client-facing wording only)

### **4. Actual Output (Observed Behavior)**

Document what the engine *actually* displayed:

- UI behavior
- Graphs/charts
- Flags
- Summary text
- Any anomalies or mismatches

### **5. Alignment Check (Expected vs Actual)**

For each scenario:

- Fully aligned
- Partially aligned
- Misaligned
- Missing output
- Unexpected output
- UI rendering issues
- Scoring inconsistencies

### **6. Client-Facing Clarity Check**

Evaluate whether the output is:

- Clear
- Non-technical
- Actionable
- Free of internal logic exposure
- Consistent with your brand voice
- Appropriate for executive-level review

## **7. Internal Logic Exposure Check**

Confirm the engine does **NOT** expose:

- SPC formulas
- Capability indices
- DOE triggers
- Internal scoring logic
- Root cause inference
- Diagnostic pathways
- Constraint analysis
- Any engineering terminology not meant for clients

## **8. Bugs, Issues, or Required Adjustments**

Document:

- UI bugs
- Scoring errors
- Incorrect flags
- Missing summaries
- Misleading client-facing language
- Any internal logic that leaked
- Any required adjustments for Cursor